

EE2100: Matrix Theory**Exam - 1****Name:** _____**Roll/Reg. No.:** _____**Instructions**

1. The exam is for 90 minutes. It will commence at 11:10. You must hand over the question paper and the answer book before 12:40. Please note **any delay** in submission may attract **negative marking**.
2. Go through all the questions carefully. Any misinterpretations will NOT be considered.
3. If you feel any question is incomplete, you can complete the solution by assuming appropriate values for the missing information.
4. If you feel that in any question, the operation is not defined because the operands on either side are not compatible, write that as the solution.
5. The paper will have three sections. Section I contains simple questions for which you should write answers (for questions 1 and 2) in the space provided adjacent to the question. Note that answers written elsewhere will NOT be considered for evaluation.
6. The solutions to questions in Sections II and III should be written in the answer book.
7. For True or False questions, clearly write "True/False." Any other form of answer, such as (T/F, etc.) won't be considered.
8. You must hand over the question paper and the answer book to the TA.

Section I (Subtotal: 10 Points)

1. (6 Points) State if the following statements are True/False.
 - (a) The zero vector (or the additive identity) is always an element of any subspace of a vector space. _____
 - (b) The scalar projection of $\mathbf{a} \in \mathcal{R}^N$ onto $\mathbf{b} \in \mathcal{R}^N$ equal to scalar projection of $\mathbf{b}$ onto $\mathbf{a}$. _____
 - (c) The vector $\mathbf{v} = [v_1, v_2, 0] \in \mathcal{R}^3$, where $v_1, v_2 \in \mathcal{R}$, with usual vector addition and scalar multiplication is a subspace of $\mathcal{R}^3$. _____
 - (d) Consider three vectors $\mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathcal{R}^N$. If the inner product of the following vector pairs $\mathbf{a} \cdot \mathbf{b}$ and $\mathbf{a} \cdot \mathbf{c}$ is equal and non-zero for a given $\mathbf{a} \in \mathcal{R}^N$, i.e., $\mathbf{a} \cdot \mathbf{b} = \mathbf{a} \cdot \mathbf{c} \neq 0$, then $\mathbf{b} = \mathbf{c}$. _____
 - (e) Let $\mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathcal{R}^N$ such that $\mathbf{c} = \alpha_1 \mathbf{a} + \alpha_2 \mathbf{b}$ (where $\alpha_1, \alpha_2 \in \mathcal{R}$). Then $\text{Span}\{\mathbf{a}, \mathbf{b}, \mathbf{c}\} = \text{Span}\{\mathbf{a}, \mathbf{b}\}$. _____
 - (f) The basis of any subspace cannot contain the zero vector of the corresponding subspace. _____

2. (2 points) (Indicate Yes/No with suitable arguments) Does the following definition for the norm of a vector $\mathbf{u} \in \mathcal{R}^N$

$$\|\mathbf{u}\| = \min_i |u_i| \quad (1)$$

satisfy all the properties required for the norm?. _____

3. (2 Points) Consider $\mathbf{e}_x$ and $\mathbf{e}_y$ (unit vectors) denote the directions of $\mathbf{x} \in \mathcal{R}^N$ and $\mathbf{y} \in \mathcal{R}^N$ respectively. If $\mathbf{x} \perp \mathbf{y}$, what is the value of $\|\mathbf{e}_x + \mathbf{e}_y\|_2^2$ and $\|\mathbf{e}_x - \mathbf{e}_y\|_2^2$.

Section II (Subtotal: 10 Points)

1. (3 Points) Do the vectors $\mathbf{x} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\mathbf{y} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ constitute a basis for $\mathcal{R}^2$. If so, compute the coordinate vector of $\mathbf{z}$, which in the standard basis is given by $\mathbf{z} = [0, 1]$.
2. (3 Points) Consider two vectors $\mathbf{a}, \mathbf{b} \in \mathcal{R}^N$. Show that the vector $\mathbf{b} - \text{Proj}_{\mathbf{a}} \mathbf{b}$ is orthogonal to $\mathbf{a}$. Make sure to write down all the finer details.
3. (4 Points) Consider the set of vectors $\mathbb{V} = \{\mathbf{v}_1, \dots, \mathbf{v}_m\} \in \mathcal{R}^2$. The vector addition and scalar multiplication are defined as

$$\mathbf{v}_i + \mathbf{v}_j = \begin{bmatrix} v_{i1} + v_{j1} \\ 0 \end{bmatrix} \text{ and } \alpha \cdot \mathbf{v}_i = \begin{bmatrix} \alpha v_{i1} \\ \alpha v_{i2} \end{bmatrix} \quad (2)$$

Do the set of vectors defined over the field of real numbers together with the defined operators constitute a vector space? State the necessary arguments.

Section III (Subtotal: 10 Points)

1. (4 Points) Consider a vector $\mathbf{u}$ in subspace $\mathbb{W}$ whose orthogonal basis is $\{\mathbf{b}_1, \dots, \mathbf{b}_m\}$. If $\mathbf{u}_\beta$ (given by Eq. (3)) is the coordinate vector of $\mathbf{u}$ in the orthogonal basis, compute the coordinates of vector $\mathbf{u}$ on the standard basis.

$$\mathbf{u}_\beta = \begin{bmatrix} u_{\beta 1} \\ \vdots \\ u_{\beta m} \end{bmatrix} \quad (3)$$

2. (6 Points) Prove or Disprove with necessary arguments/proofs. Let $\{\mathbf{u}_1, \dots, \mathbf{u}_m\}$ denote a collection of linearly independent vectors in the standard basis of a subspace $\mathbb{W}$. If $\mathbb{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_m\}$ denote any basis (except the standard basis) of the subspace $\mathbb{W}$, then, the collection of vectors $\{\mathbf{u}_{\beta 1}, \dots, \mathbf{u}_{\beta m}\}$ (where $\mathbf{u}_{\beta j}$ is the coordinate vector of $\mathbf{u}_j$ in basis $\mathbb{B}$), are linearly independent.